

RAM-ADMM based DoA Estimation for semi-passive IRS-aided sensing system

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Abstract—In this article, a super-resolution direction-of-arrival (DoA) estimation framework for non-line-of-sight (NLoS) target localization in semi-passive Intelligent reflecting surface (IRS)-aided sensing systems has been presented. In this proposed scheme, reweighted atomic norm minimization (RAM) and virtual array construction are integrated. By utilizing the log-determinant function to promote low-rank properties more aggressively than the conventional nuclear norm, a received signal recovery problem is formulated. With the recovered received signal, an efficient Alternating Direction Method of Multipliers (ADMM)-based solver is employed to alleviate the computational burden, and the final DoA estimates are retrieved via Hankel-based Root-MUSIC. Furthermore, it avoids the high computational burden from interior-point methods, and achieves a desirable balance between accuracy and computational complexity. Numerical results verify that the proposed RAM-ADMM framework achieves accuracy comparable to ANM while significantly reducing computational complexity.

Index Terms—Intelligent reflecting surface (IRS), direction of arrival (DoA), non-line-of-sight (NLoS) target localization, reweighted atomic norm minimization (RAM), Alternating Direction Method of Multipliers (ADMM).

I. INTRODUCTION

Direction-of-arrival (DoA) estimation is fundamental to radar sensing. While methods based on compressed sensing [1], tensor decomposition [2]–[4], and deep learning [5]–[7] have achieved remarkable precision, their performance degrades severely in non-line-of-sight (NLoS) scenarios where physical obstacles block direct signal paths. Since such blockages are prevalent in urban environments, purely algorithmic advances cannot fully address this challenge. Intelligent reflecting surfaces (IRS) offer a hardware-level solution by reconfiguring the propagation environment to establish virtual line-of-sight links [8].

This work was supported in part by Hainan Province Science and Technology Special Fund under Grant ZDYF2023GXJS168, ZDYF2023GXJS159 and ZDYF2025SHFZ058, in part by the National Natural Science Foundation of China(62461020,62101165,62501201), in part by the Scientific research projects of higher education institutions in Hainan Province under Grant Hnky2025-1.

Existing IRS-aided sensing research has predominantly focused on fully-passive architectures. A representative line of work by Chen *et al.* demonstrated that DoA estimation can be achieved with only a single receiving channel by exploiting temporal phase variations at the IRS, where atomic norm minimization (ANM) and its efficient alternating direction method of multipliers (ADMM)-based solver were developed to leverage target sparsity in the spatial domain [9]–[11]. In parallel, bistatic MIMO radar systems equipped with forward-backward IRS configurations have been investigated for joint direction-of-departure (DoD) and DoA estimation in NLoS scenarios, with subspace methods such as reduced-dimension MUSIC and higher-order propagator methods proposed to reduce computational complexity [12], [13]. However, the inherent double-reflection path loss in fully-passive structures severely limits sensing range. To mitigate this, the semi-passive IRS architecture—integrating Active Sensing Elements (ASEs) with passive reflectors—emerges as an ideal trade-off between hardware complexity and signal strength. Unlocking the full potential of semi-passive systems, however, requires distinct algorithmic innovation [14]–[16].

In this paper, we develop a reweighted atomic norm minimization (RAM) [17] framework for DoA estimation in semi-passive IRS-aided sensing systems. To overcome the resolution loss caused by convex relaxation in conventional atomic norm minimization, we employ the log-determinant function as a tighter nonconvex surrogate for the rank metric and solve the resulting problem via a majorization-minimization procedure. An ADMM-based solver is further developed to reduce the per-iteration complexity that dominated by eigenvalue decomposition. Finally, a virtual uniform linear array is constructed through redundancy averaging and applies a Hankel-based Root-MUSIC estimator to achieve super-resolution DoA retrieval.

II. SEMI-PASSIVE IRS-AIDED SENSING SYSTEM MODEL

Investigated in [15], the semi-passive IRS architecture is adopted to mitigate the severe path loss associated with fully

passive IRS schemes. The system comprises a Base Station (BS), Passive Reflecting Elements (PREs), and Active Sensing Elements (ASEs), which are equipped with Uniform Linear Arrays (ULAs) consisting of Q_B , Q_P , and Q_A elements, respectively. To localize K far-field targets in the Non-Line-of-Sight (NLoS) region, we utilize the signal model derived in [15]. Consequently, the received signal, stacked over T snapshots, is expressed as:

$$\mathbf{Z} = [\mathbf{z}(1), \mathbf{z}(2), \dots, \mathbf{z}(T)] \\ = \mathbf{A}_A(\boldsymbol{\theta}) \mathbf{\Lambda} \mathbf{A}_P^H(\bar{\boldsymbol{\theta}}) \mathbf{S} + \mathbf{N} \in \mathbb{C}^{Q_A \times T}, \quad (1)$$

where $\mathbf{A}_A(\boldsymbol{\theta}) = [\mathbf{a}_A(\theta_1), \dots, \mathbf{a}_A(\theta_K)] \in \mathbb{C}^{Q_A \times K}$ and $\mathbf{A}_P(\bar{\boldsymbol{\theta}}) = [\mathbf{a}_P(\bar{\theta}_1), \dots, \mathbf{a}_P(\bar{\theta}_K)] \in \mathbb{C}^{Q_P \times K}$ denote the array manifold matrices for the ASEs and PREs, respectively. Here, θ_k denotes the DoA of the k -th target relative to the ASEs. The equivalent angle for the PREs is given by $\bar{\theta}_k = \arcsin(\sin(\theta_k) - \sin(\theta_{B2P}))$, where θ_{B2P} represents the known angle of departure from the BS to the PREs. The diagonal matrix $\mathbf{\Lambda} = \text{diag}(\beta_1, \dots, \beta_K) \in \mathbb{C}^{K \times K}$ captures the composite channel gain. Assuming perfect beam alignment between the BS and PREs, the complex coefficient for the k -th target is defined as:

$$\beta_k = \sqrt{Q_B} \sigma_t \alpha_{B2P} \alpha_k \\ = \sqrt{Q_B} \sigma_t \left(\frac{\lambda}{4\pi d_{B2P}} e^{-j\frac{2\pi d_{B2P}}{\lambda}} \right) \left(\frac{\lambda \sqrt{\gamma_k}}{8\pi^{1.5} d_k^2} e^{-j\frac{4\pi d_k}{\lambda}} \right), \quad (2)$$

where σ_t is the transmit signal amplitude, λ is the carrier wavelength, and γ_k represents the Radar Cross Section (RCS) of the k -th target. d_{B2P} and d_k denote the distances from the BS to the IRS and from the IRS to the k -th target, respectively. The pilot matrix $\mathbf{S} = [\phi(1)d(1), \dots, \phi(T)d(T)] \in \mathbb{C}^{Q_P \times T}$ is presumed to be pre-configured and synchronized with the IRS controller. Here, $\phi(t) = [e^{j\phi_1(t)}, e^{j\phi_2(t)}, \dots, e^{j\phi_{Q_P}(t)}]^T$ denotes the phase shift vector, and $d(t)$ is the unit-power pilot symbol. For analytical tractability, $\mathbf{S}$ is designed to satisfy the orthogonality property $\frac{1}{T} \mathbf{S} \mathbf{S}^H = \mathbf{I}_{Q_P}$. $\mathbf{N}$ is the additive white Gaussian noise matrix, with independent and identically distributed entries following $\mathcal{CN}(0, \sigma_n^2)$, where σ_n^2 denotes the noise power. Assuming half-wavelength element spacing, the steering vector for a generic angle ϑ with Q elements is defined as

$$\mathbf{a}(\vartheta) \triangleq [1, e^{j\pi \sin \vartheta}, \dots, e^{j(Q-1)\pi \sin \vartheta}]^T \in \mathbb{C}^{Q \times 1}. \quad (3)$$

III. PROPOSED RAM-ADMM ALGORITHM FOR TARGET LOCALIZATION

A. Reweighted Atomic Norm Minimization Formulation

To recover the low-rank channel matrix $\mathbf{X} \in \mathbb{C}^{Q_A \times Q_P}$ from the noisy observations $\mathbf{Z}$ described in (1), we leverage the atomic norm minimization framework. While the standard nuclear norm serves as a convex surrogate for the rank function, it often yields suboptimal sparsity under limited snapshots or low SNR conditions. To enhance the resolution of target localization, we adopt the log-determinant function $\log |\cdot|$ as a tighter nonconvex relaxation of the rank metric. Specifically,

the recovery problem is formulated as a reweighted Semi-Definite Programming (SDP) problem:

$$\min_{\mathbf{X}, \mathbf{u}_1, \mathbf{u}_2} \quad \eta (\text{tr}(\mathbf{W}_1 \mathbf{T}(\mathbf{u}_1)) + \text{tr}(\mathbf{W}_2 \mathbf{T}(\mathbf{u}_2))) + \frac{1}{2} \|\mathbf{Z} - \mathbf{X} \mathbf{S}\|_F^2 \\ \text{s.t.} \quad \begin{bmatrix} \mathbf{T}(\mathbf{u}_1) & \mathbf{X} \\ \mathbf{X}^H & \mathbf{T}(\mathbf{u}_2) \end{bmatrix} \succeq 0 \quad (4)$$

where $\mathbf{T}(\mathbf{u}_1)$ denotes a Hermitian Toeplitz matrix whose first column is determined by the vector $\mathbf{u}_1$, and $\mathbf{T}(\mathbf{u}_2)$ is defined analogously with $\mathbf{u}_2$. The weight constant is defined as $\eta \triangleq \frac{\rho}{2\sqrt{Q_A Q_P}}$, where ρ is a regularization parameter.

To address the nonconvexity, we employ the Majorization-Minimization (MM) strategy. In the l -th iteration, the weighting matrices $\mathbf{W}_1$ and $\mathbf{W}_2$ are updated based on the estimates from the previous iteration:

$$\mathbf{W}_1^{(l)} = (\mathbf{T}(\mathbf{u}_1)^{(l-1)} + \epsilon \mathbf{I}_{Q_A})^{-1}, \quad (5)$$

$$\mathbf{W}_2^{(l)} = (\mathbf{T}(\mathbf{u}_2)^{(l-1)} + \epsilon \mathbf{I}_{Q_P})^{-1}, \quad (6)$$

where $\epsilon > 0$ is a small smoothing constant to ensure numerical stability.

B. Efficient Solver based on ADMM

Solving problem (4) via standard interior-point methods (e.g., SDPT3 or SeDuMi) incurs a prohibitive computational complexity of $\mathcal{O}((Q_A + Q_P)^{3.5})$ per iteration. To expedite the sensing process, we develop an efficient solver based on the Alternating Direction Method of Multipliers (ADMM).

We first reshape the constraints by introducing a block auxiliary variable $\mathcal{M} = [\mathbf{T}(\mathbf{u}_1), \mathbf{X}; \mathbf{X}^H, \mathbf{T}(\mathbf{u}_2)]$ and a slack variable $\mathcal{P}$. The Augmented Lagrangian function is constructed as:

$$\mathcal{L}_\mu(\mathcal{M}, \mathcal{P}, \mathbf{\Lambda}) = \eta \text{tr}(\mathbf{W}_1 \mathbf{T}(\mathbf{u}_1)) + \eta \text{tr}(\mathbf{W}_2 \mathbf{T}(\mathbf{u}_2)) \\ + \frac{1}{2} \|\mathbf{Z} - \mathbf{X} \mathbf{S}\|_F^2 + \langle \mathbf{\Lambda}, \mathcal{P} - \mathcal{M} \rangle \quad (7) \\ + \frac{\mu}{2} \|\mathcal{P} - \mathcal{M}\|_F^2 + \mathbb{I}_{\mathcal{S}_+}(\mathcal{P}),$$

where $\mathbf{\Lambda}$ is the dual multiplier matrix partitioned as $\mathbf{\Lambda} = [\mathbf{\Lambda}_1, \mathbf{\Lambda}_{12}; \mathbf{\Lambda}_{12}^H, \mathbf{\Lambda}_2]$, μ is the penalty parameter, and $\mathbb{I}_{\mathcal{S}_+}(\cdot)$ is the indicator function for the positive semi-definite cone.

The ADMM algorithm alternates between updating $\mathcal{M}$, $\mathcal{P}$, and $\mathbf{\Lambda}$. The closed-form updates for the i -th inner iteration are derived as follows:

1) *Primal Variable Update ($\mathcal{M}$):* The update for $\mathbf{X}$ is a regularized least-squares problem:

$$\mathbf{X}^{(i+1)} = \left(\mathbf{Z} \mathbf{S}^H + 2\mu \tilde{\mathcal{P}}_{12}^{(i)} \right) \left(\mathbf{S} \mathbf{S}^H + 2\mu \mathbf{I}_{Q_P} \right)^{-1}, \quad (8)$$

where $\tilde{\mathcal{P}}_{12}^{(i)} = \mathcal{P}_{12}^{(i)} + \mu^{-1} \mathbf{\Lambda}_{12}^{(i)}$. Note that utilizing the orthogonality property $\frac{1}{T} \mathbf{S} \mathbf{S}^H = \mathbf{I}$ derived in Section II simplifies the matrix inversion significantly.

The updates for the Hermitian Toeplitz matrices $\mathbf{T}(\mathbf{u}_1)$ and $\mathbf{T}(\mathbf{u}_2)$ involve projections onto the Hermitian Toeplitz set $\Pi_{\mathcal{T}^H}(\cdot)$:

$$\begin{aligned}\mathbf{T}(\mathbf{u}_1^{(i+1)}) &= \Pi_{\mathcal{T}^H} \left(\mathcal{P}_1^{(i)} + \mu^{-1} \mathbf{\Lambda}_1^{(i)} - \mu^{-1} \eta \mathbf{W}_1 \right), \\ \mathbf{T}(\mathbf{u}_2^{(i+1)}) &= \Pi_{\mathcal{T}^H} \left(\mathcal{P}_2^{(i)} + \mu^{-1} \mathbf{\Lambda}_2^{(i)} - \mu^{-1} \eta \mathbf{W}_2 \right).\end{aligned}\quad (9)$$

2) *Auxiliary Variable Update ($\mathcal{P}$)*: The variable $\mathcal{P}$ is updated by projecting onto the PSD cone. Let $\mathbf{Q}^{(i+1)} = \mathbf{M}^{(i+1)} - \mu^{-1} \mathbf{\Lambda}^{(i)}$. By performing the EVD $\mathbf{Q}^{(i+1)} = \mathbf{E} \mathbf{\Sigma} \mathbf{E}^H$, the update is given by

$$\mathcal{P}^{(i+1)} = \Pi_{\mathcal{S}_+} \left(\mathbf{M}^{(i+1)} - \mu^{-1} \mathbf{\Lambda}^{(i)} \right) = \mathbf{E} \max(\mathbf{\Sigma}, \mathbf{0}) \mathbf{E}^H. \quad (10)$$

3) *Dual Update ($\mathbf{\Lambda}$)*: Finally, the dual variables are updated via gradient ascent:

$$\mathbf{\Lambda}^{(i+1)} = \mathbf{\Lambda}^{(i)} + \mu (\mathcal{P}^{(i+1)} - \mathbf{M}^{(i+1)}). \quad (11)$$

This ADMM procedure iterates until the primal and dual residuals fall below a predefined tolerance.

C. Target Parameter Estimation Strategy

Given the optimal low-rank matrix $\mathbf{X}^*$ obtained from the proposed optimization algorithm, we exploit its structural properties to estimate the DoAs with enhanced resolution. The estimation procedure involves three main stages: phase compensation, virtual array construction, and subspace-based retrieval.

Let the recovered matrix element at the (m, n) -th position be denoted as $[\mathbf{X}^*]_{m,n}$. Ignoring the noise term, it inherently approximates the ideal cascaded channel $\mathbf{A}_A \mathbf{\Lambda} \mathbf{A}_P^H$. Specifically, its elements can be expressed as:

$$[\mathbf{X}^*]_{m,n} \approx \sum_{k=1}^K \beta_k e^{j\pi(m-1)\sin\theta_k} e^{-j\pi(n-1)\sin\bar{\theta}_k}. \quad (12)$$

By substituting the geometric equivalent angle $\sin\bar{\theta}_k = \sin\theta_k - \sin\theta_{B2P}$ into the above equation, it becomes evident that the captured signal contains a phase coupling between the unknown DoA θ_k and the known geometric angle θ_{B2P} . To decouple them and align the array manifolds, we apply a deterministic phase compensation matrix $\mathbf{\Phi}_{\text{comp}} \in \mathbb{C}^{Q_A \times Q_P}$:

$$\tilde{\mathbf{X}} = \mathbf{X}^* \oplus \mathbf{\Phi}_{\text{comp}}, \quad \text{with } [\mathbf{\Phi}_{\text{comp}}]_{m,n} = e^{-j\pi(n-1)\sin\theta_{B2P}}, \quad (13)$$

where $\oplus$ denotes the Hadamard product.

Subsequently, the compensated element becomes $[\tilde{\mathbf{X}}]_{m,n} \approx \sum_{k=1}^K \beta_k e^{j\pi(m-n)\sin\theta_k}$. We observe that the compensated matrix $\tilde{\mathbf{X}}$ exhibits a Toeplitz-like structure, dependent purely on the spatial lag $\ell = m - n$. This enables us to map the matrices into a difference co-array, thereby constructing a virtual ULA with an extended aperture of $L_{\text{virtual}} = Q_A + Q_P - 1$. To enhance the Signal-to-Noise Ratio (SNR) of the virtual array, the virtual signal vector $\mathbf{v} \in \mathbb{C}^{L_{\text{virtual}}}$ is obtained by averaging the redundant entries of $\tilde{\mathbf{X}}$ corresponding to the same lag ℓ :

$$v(p) = \frac{1}{|\mathcal{K}_p|} \sum_{(m,n) \in \mathcal{K}_p} [\tilde{\mathbf{X}}]_{m,n}, \quad p = 1, \dots, L_{\text{virtual}}, \quad (14)$$

where the index set mapping the lag to the linear index is defined as $\mathcal{K}_p = \{(m, n) \mid (m-1) - (n-1) + Q_P = p\}$.

Since the virtual signal vector $\mathbf{v}$ essentially corresponds to a single coherent snapshot on the virtual array, direct application of subspace methods is infeasible due to the rank deficiency of its covariance matrix. To resolve this, spatial smoothing via Hankel matrix reconstruction is adopted to restore the rank structure. A Hankel matrix $\mathbf{H} \in \mathbb{C}^{L_1 \times L_2}$ is constructed from $\mathbf{v}$, where $L_1 = \lfloor L_{\text{virtual}}/2 \rfloor + 1$ and $L_2 = L_{\text{virtual}} - L_1 + 1$:

$$\mathbf{H} = \mathcal{H}(\mathbf{v}) = \begin{bmatrix} v(1) & v(2) & \dots & v(L_2) \\ v(2) & v(3) & \dots & v(L_2 + 1) \\ \vdots & \vdots & \ddots & \vdots \\ v(L_1) & v(L_1 + 1) & \dots & v(L_{\text{virtual}}) \end{bmatrix}. \quad (15)$$

Finally, singular value decomposition (SVD) is applied to matrix $\mathbf{H}$ to obtain the noise subspace $\mathbf{U}_n$. Based on $\mathbf{U}_n$, the DoAs are retrieved via the standard Root-MUSIC procedure by selecting the K roots closest to the unit circle, and the corresponding angles are computed from the root phases.

Algorithm 1 Proposed RAM-ADMM for IRS-Aided Target Localization

Require: Observation received signal $\mathbf{Z}$, pilot matrix $\mathbf{S}$, parameters ρ, μ, ϵ , maximum MM iterations L_{max} , maximum ADMM iterations I_{max} , tolerance δ .

Ensure: Target DoA estimates $\{\hat{\theta}_k\}_{k=1}^K$.

- 1: **Initialization:** Set MM iteration $l = 0$. Initialize $\mathbf{u}_1^{(0)}, \mathbf{u}_2^{(0)}$ with zeros.
 - 2: **repeat**
 - 3: $l \leftarrow l + 1$
 - 4: Update weight matrices $\mathbf{W}_1^{(l)}$ and $\mathbf{W}_2^{(l)}$ via Eq. (5) and (6).
 - 5: **Initialize ADMM:** $i = 0, \mathcal{M}^{(0)} = \mathbf{0}, \mathcal{P}^{(0)} = \mathbf{0}, \mathbf{\Lambda}^{(0)} = \mathbf{0}$.
 - 6: **repeat**
 - 7: $i \leftarrow i + 1$
 - 8: Update primal variable $\mathbf{X}^{(i)}$ via Eq. (8).
 - 9: Update Toeplitz matrices $\mathbf{T}(\mathbf{u}_1^{(i)})$ and $\mathbf{T}(\mathbf{u}_2^{(i)})$ via Eq. (9).
 - 10: Update auxiliary variable $\mathcal{P}^{(i)}$ via EVD thresholding in Eq. (10).
 - 11: Update dual multiplier $\mathbf{\Lambda}^{(i)}$ via Eq. (11).
 - 12: **until** primal and dual residuals fall below δ or $i \geq I_{\text{max}}$
 - 13: $\mathbf{X}^* \leftarrow \mathbf{X}^{(i)}, (\mathbf{u}_1^{(l)}, \mathbf{u}_2^{(l)}) \leftarrow (\mathbf{u}_1^{(i)}, \mathbf{u}_2^{(i)})$.
 - 14: **until** convergence or $l \geq L_{\text{max}}$
 - 15: Compensate phase $\tilde{\mathbf{X}} = \mathbf{X}^* \oplus \mathbf{\Phi}_{\text{comp}}$ using θ_{B2P} .
 - 16: Construct virtual signal $\mathbf{v}$ by averaging redundant lags via Eq. (14).
 - 17: Form Hankel matrix $\mathbf{H} = \mathcal{H}(\mathbf{v})$ and compute its SVD.
 - 18: Retrieve $\{\hat{\theta}_k\}_{k=1}^K$ using Root-MUSIC based on the noise subspace of $\mathbf{H}$.
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To provide a clear overview of the implementation, the complete procedure of the proposed target localization method is summarized in **Algorithm 1**.

IV. SIMULATION RESULT

This section presents several Monte Carlo experiments designed to evaluate the performance of the proposed method. Unless otherwise specified, the experimental parameters are set as follows: $Q_B = Q_P = 48$, $Q_A = 10$, $T = 400$, the carrier frequency is set to 28GHz, $d(t) = 1$, $d_{B2P} = 40\text{m}$, $d_k = 5\text{m}$, $\gamma_k = 10\text{dBsm}$, The noise power $\sigma_n^2 = -110\text{dBm}$. There are $K = 3$ far-field targets $(\theta_1, \theta_2, \theta_3) = (-10^\circ, 10^\circ, 30^\circ)$. The DFT codebook matrix $\mathbf{S}$ is set as the same as [15]. The algorithms in [14], [15], and [16] are denoted as MUSIC, WESPRIT, and ANM, respectively, and the RCRB [16] is also included as an important benchmark. To ensure consistency, the subsequent parameter estimation strategy for ANM is kept the same as in this paper, instead of directly adopting the strategy in [16]. Before starting the first numerical simulation, root mean square error (RMSE), which is defined as

$$\text{RMSE} = \sqrt{\frac{1}{K \cdot 500} \sum_{t=mc=1}^{500} \sum_{k=1}^K \left| \hat{\theta}_{k,t} - \theta_k \right|^2}.$$

Fig. 1 plots the estimation accuracy against transmit power σ_t^2 . It is observed that WESPRIT yields lower errors than MUSIC due to its utilization of the full array aperture ($\mathbf{A}_A$ and $\mathbf{A}_P$). Notably, the proposed method matches the performance of the benchmark ANM while significantly outperforming both MUSIC and WESPRIT. This performance gain validates that our approach successfully exploits both the increased degrees of freedom from the virtual array and the signal sparsity, avoiding the limitations of subspace-based techniques.

Fig. 2 and Fig. 3 investigate the impact of the number of PREs on estimation accuracy and computational efficiency, respectively. In terms of accuracy (Fig. 2), the RMSE of WESPRIT gradually decreasing and surpassing MUSIC as the number of PREs increases. This confirms that WESPRIT effectively utilizes the extended aperture provided by the additional PREs, whereas MUSIC is primarily limited by the fixed active array. Meanwhile, the proposed method achieves the lowest RMSE, comparable to ANM, by fully exploiting the sparsity of the virtual array signals. However, the computational advantage of the proposed method is explicitly highlighted in Fig. 3. While ANM suffers from a prohibitive computational burden due to the heavy SDP operations, the proposed method significantly reduces the runtime, matching the efficiency of subspace-based algorithms (MUSIC and WESPRIT). Unlike ANM, which sacrifices speed for accuracy, our approach successfully strikes a desirable balance, achieving high-super-resolution performance with low computational complexity.

V. CONCLUSION

In this paper, a computationally efficient gridless DoA estimation framework using semi-passive IRS-aided sensing systems has been presented. In this algorithm, a RAM strategy is applied, and the virtual array statistics are leveraged. With the virtual array and Hankel matrix constructed, the DoA estimates are obtained via Root-MUSIC. It significantly reduces the computational burden compared to ANM through using ADMM solver. Numerical results demonstrate that the

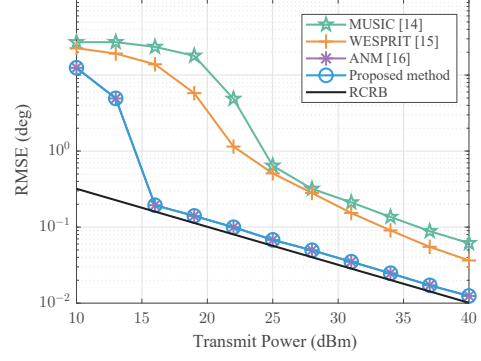


Fig. 1. Comparison of RMSE under varying transmit power σ_t^2 levels.

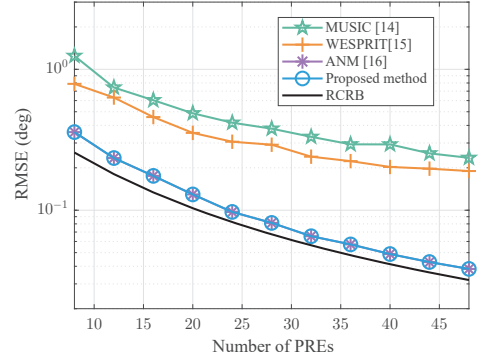


Fig. 2. Comparison of RMSE under varying Number of PREs.

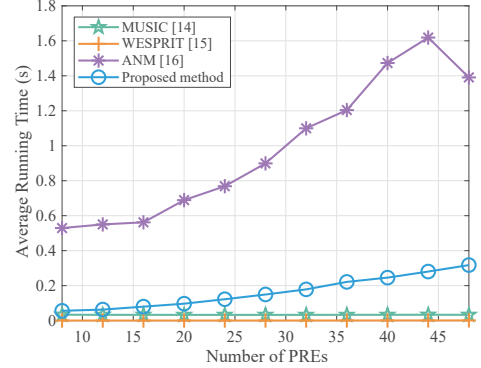


Fig. 3. Comparison of Average running time under varying Number of PREs.

proposed algorithm achieves estimation accuracy comparable to ANM—while significantly outperforming other subspace-based benchmarks—and offers a substantial advantage in terms of runtime efficiency. Thus, it strikes a desirable balance between high-resolution sensing and computational complexity.

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